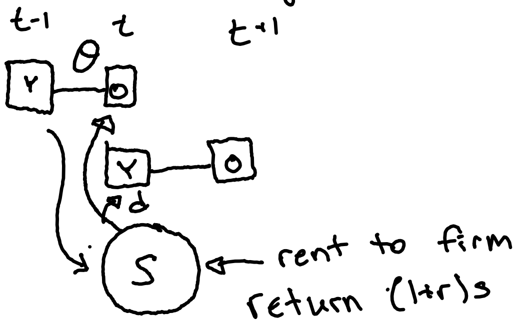
$\boxed{04\text{-}02\text{-}2025}$

Model w/ Accidental Bequests in OLG:

↳ accidental bequest ⟹ survival probability



↳ eq. for bequest:

$L_t \cdot d :=$ bequest received

$$L_t \cdot d = \underbrace{L_{t-1}(1-\theta)s(1+r)}_{\text{total remaining bequest}}.$$

$$\boxed{d = \frac{1}{1+n} \cdot (1-\theta)s(1+r)}$$

Firm:

$\delta = 1$, $1+r = \alpha k^{\alpha-1}$, $w = (1-\alpha)k^{\alpha}$

Capital:

$s = k(1+n)$

Consumer:

$$\max_{c^y, c^o, s} \ln c^y + \beta\theta \ln c^o$$

s.t.

$c^y = w + d - s$

$c^o = s(1+r) \Rightarrow d$ is exog. to consumer

$$\Rightarrow c^y + \frac{c^o}{1+r} = w + d$$

$$\underline{c^y = \frac{1}{1+\beta\theta}(w+d)}, \quad s = w + d - c^y = w + d - \frac{1}{1+\beta\theta}(w+d)$$

$$\underline{s = \frac{\beta\theta}{1+\beta\theta}(w+d)}$$

$$s = \frac{\beta\theta}{1+\beta\theta}\left(w + \frac{(1-\theta)s(1+r)}{1+n}\right)$$

$\Rightarrow$ only after solving con. prob.

$$s(1+\beta\theta) = \beta\theta w + \frac{\beta\theta(1-\theta)s(1+r)}{1+n}$$

$$s\left(1 + \beta\theta - \frac{\beta\theta(1-\theta)(1+r)}{1+n}\right)$$

$$= s\left(\frac{(1+\beta\theta)(1+n) - \beta\theta(1-\theta)(1+r)}{1+n}\right) = \beta\theta w$$

$$s\left(\frac{(1+\beta\theta)(1+n)-\beta\theta(1-\theta)(1+r)}{1+n}\right)=\beta\theta w$$

$$s=\frac{\beta\theta w(1+n)}{(1+\beta\theta)(1+n)-\beta\theta(1-\theta)(1+r)}$$

↳ plug into MCC for c.m.

$$k(1+n)=\frac{\beta\theta(1-\alpha)k^{\alpha}(1+n)}{(1+\beta\theta)(1+n)-\beta\theta(1-\theta)(\alpha k^{\alpha-1})}$$

$$k=\frac{\beta\theta(1-\alpha)k^{\alpha}}{(1+\beta\theta)(1+n)-\beta\theta(1-\theta)(\alpha k^{\alpha-1})}$$

$$k^{1-\alpha}=\frac{\beta\theta(1-\alpha)}{(1+\beta\theta)(1+n)-\beta\theta(1-\theta)(\alpha k^{\alpha-1})}$$

$$\beta\theta(1-\alpha)=(1+\beta\theta)(1+n)k^{1-\alpha}-\beta\theta(1-\theta)\alpha$$

$$(1+\beta\theta)(1+n)k^{1-\alpha}=\beta\theta(1-\alpha)-\beta\theta(1-\theta)\alpha$$

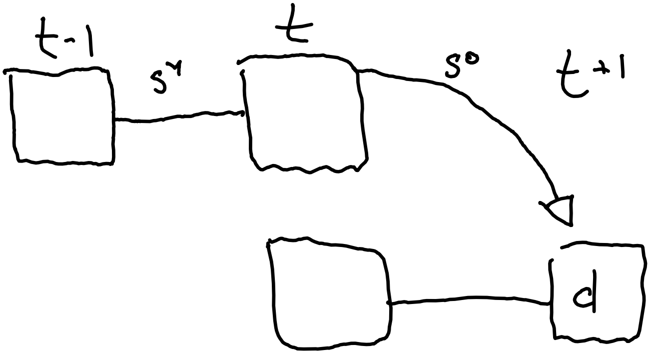
$$k=\left[\frac{\beta\theta(1-\alpha)-\beta\theta(1-\theta)\alpha}{(1+\beta\theta)(1+n)}\right]^{1/1-\alpha}$$

To check if doing right, set $\theta = 1$ for baseline

$$k = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}\right]^{1/1-\alpha}$$

Intentional Bequest:

$\theta = 1$



① bequest goes to old descendants

② bequest is intentional

Equilibrium Bequests:

$$dL_t = s^o(1+r)\,L_{t-1}$$

$$d = \frac{s^o(1+r)}{1+n}$$

Capital Market:

$$K_{t+2} = L_{t+1}s^y + L_t s^o$$

$$s^y + \frac{s^o}{(1+n)} = k(1+n)$$

Consumer's Problem:

$$V^y = \ln(c^y) + \beta\ln(c^o) + \beta^2\eta\ln(beq)$$

s.t.

$$c^y = w - s^y$$

$$c^o = d + s^y(1+r) - s^o$$

$$beq = s^o(1+r)$$

$$V^o = \ln(c^o) + \beta\eta\ln(s^o(1+r))$$

s.t.

$$c^o = \underbrace{d + s^y(1+r)}_{I} - s^o$$

$$V^o = \ln(I - s^o) + \beta\eta\ln(s^o) + cons$$

$$\frac{1}{I-s^o} = \frac{\beta\eta}{s^o} \Rightarrow s^o = \frac{\beta\eta}{1+\beta\eta} I$$

$$c^o = I - s^o = I - \frac{\beta\eta}{1+\beta\eta} I = \frac{1}{1+\beta\eta} I$$

$$V^y = \ln(w-s^y) + \beta \ln\left(\frac{1}{1+\beta\eta}(d+s^y(1+r))\right) + \beta^2\eta \ln\left(\frac{\beta\eta}{1+\beta\eta}(d+s^y(1+r))(1+r)\right)$$

$$= \ln(w-s^y) + \beta\ln(d+s^y(1+r)) + \beta^2\eta\ln(d+s^y(1+r)) + \text{cons}$$

$$= \ln(w-s^y) + \beta(1+\beta\eta)\ln(d+s^y(1+r)) + \text{cons}$$

$$\frac{1}{w-s^y} = \frac{\beta(1+\beta\eta)(1+r)}{d+s^y(1+r)} \Rightarrow d+s^y(1+r) = \beta(1+\beta\eta)(1+r)w - \beta(1+\beta\eta)(1+r)s^y$$

$$s^y(1+r)\left(1+\beta(1+\beta\eta)\right) = \beta(1+\beta\eta)(1+r)w - d$$

$$s^y = \frac{\beta(1+\beta\eta)(1+r)w - d}{(1+r)\left(1+\beta(1+\beta\eta)\right)}$$

$$\eta = 0 \Rightarrow d = 0 \Rightarrow \frac{\beta(1+r)w}{(1+r)(1+\beta)} = \frac{\beta}{(1+\beta)} \cdot w$$

concepts of Equilibrium:

1. Models w/ Individual Heterogeneity: (SIM)
   → Stationary competitive equilibrium

2. Models w/ Aggregate Uncertainty
   → Recursive C.E.

Models w/ Indiv. + Agg. Uncert-y: RCE

Key features of first generation SIM:

1. No ex-ante heterogeneity
2. Infinite Horizon
3. Borrowing Constraint
4. No aggregate uncertainty
5. only consumption + saving decision
6. simple institutional framework

GE framework:

→ consump/saving ⟹ into G.E.

→ representative agent model ⟹ relax complete market assump